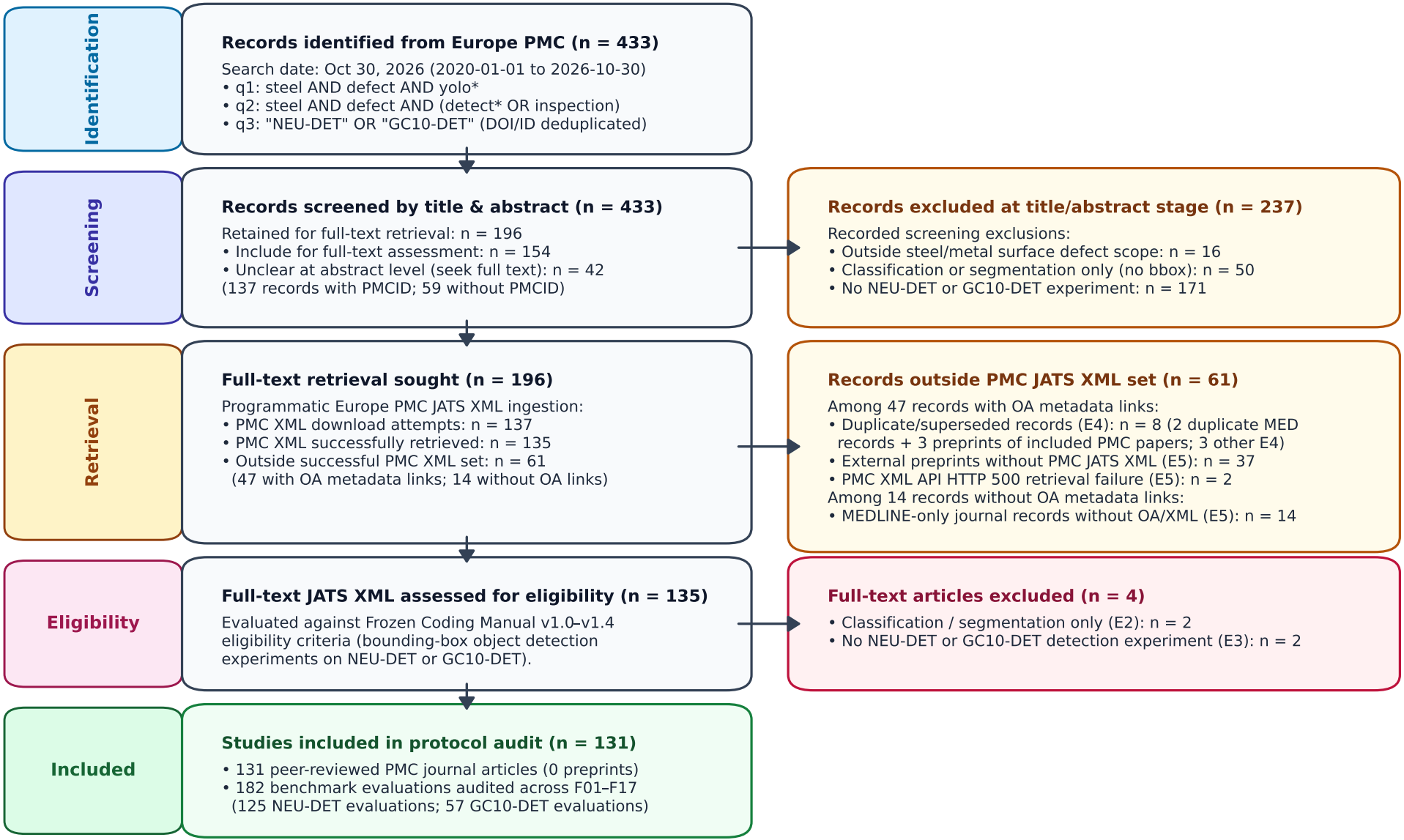
Identification
Records identified from Europe PMC (n = 433)
Search date: Oct 30, 2026 (2020-01-01 to 2026-10-30)
• q1: steel AND defect AND yolo*
• q2: steel AND defect AND (detect* OR inspection)
• q3: "NEU-DET" OR "GC10-DET" (DOI/ID deduplicated)
Screening
Records screened by title & abstract (n = 433)
Retained for full-text retrieval: n = 196
• Include for full-text assessment: n = 154
• Unclear at abstract level (seek full text): n = 42
(137 records with PMCID; 59 without PMCID)
Records excluded at title/abstract stage (n = 237)
Recorded screening exclusions:
• Outside steel/metal surface defect scope: n = 16
• Classification or segmentation only (no bbox): n = 50
• No NEU-DET or GC10-DET experiment: n = 171
Retrieval
Full-text retrieval sought (n = 196)
Programmatic Europe PMC JATS XML ingestion:
• PMC XML download attempts: n = 137
• PMC XML successfully retrieved: n = 135
• Outside successful PMC XML set: n = 61
(47 with OA metadata links; 14 without OA links)
Records outside PMC JATS XML set (n = 61)
Among 47 records with OA metadata links:
• Duplicate/superseded records (E4): n = 8 (2 duplicate MED records + 3 preprints of included PMC papers; 3 other E4)
• External preprints without PMC JATS XML (E5): n = 37
• PMC XML API HTTP 500 retrieval failure (E5): n = 2
Among 14 records without OA metadata links:
• MEDLINE-only journal records without OA/XML (E5): n = 14
Eligibility
Full-text JATS XML assessed for eligibility (n = 135)
Evaluated against Frozen Coding Manual v1.0–v1.4 eligibility criteria (bounding-box object detection experiments on NEU-DET or GC10-DET).
Full-text articles excluded (n = 4)
• Classification / segmentation only (E2): n = 2
• No NEU-DET or GC10-DET detection experiment (E3): n = 2
Included
Studies included in protocol audit (n = 131)
• 131 peer-reviewed PMC journal articles (0 preprints)
• 182 benchmark evaluations audited across F01–F17
(125 NEU-DET evaluations; 57 GC10-DET evaluations)